



University of Debrecen, Faculty of Informatics

Logic in Computer Science

Preliminary introduction of mathematical logic languages

Slides supporting the lecture

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Preliminary information

Formulae and terms

In a first-order language, we distinguish two kinds of expression:

- A *formula* represents a, sometimes parametrized, statement describing some reasoning. The most simple and parameter-free form of a formula is a propositional letter that represents a single atomic statement.
- in contrast, the *terms* are used to represent objects. The most simple form of a term is a variable or a constant symbol, but objects can also be described in a compounded way.

The structure of a zero-order language is simpler, e.g., they cannot use terms.

Preliminary information

Construction of formulae

From atomic formulae by using *logical constants* we are able to construct more complex formulae.

The following logical constants are called *connectives*:

- $\neg$ is the *negation sign*, where $\neg A$ can be read as „not A”;
- $\wedge$ is the *conjunction sign*, where $(A \wedge B)$ can be read as „A and B”,
- $\vee$ is the *disjunction sign*, where $(A \vee B)$ can be read as „A or B”,
- $\supset$ is the *implication sign*, where $(A \supset B)$ can be read as „if A then B”,
- $\equiv$ is the *sign of the material equivalence*, where $(A \equiv B)$ can be read as „A if and only if B”,

supposing that A and B represent statements (they are formulae).

Preliminary information

Quantifiers

The following logical constants are called *quantifiers*:

$\forall$ is the universal quantifier, where

$\forall xA$ can be read as „ A is true for all values of x ”;

$\exists$ is the existential quantifier, where

$\exists xA$ can be read as „*there exist some value of x so that A is true*”.

Here x is a variable and A is an arbitrary formula (that usually contains x as a parameter).

Because there are no terms and variables in a zero-order language, we are unable to use quantifiers too.

Examples of first-order languages

The Subset language

The *Subset* language is a first-order language in which we can only find one type of atomic formula. Those can be written in the form of $(x \subseteq y)$ and can be read as:

„ x is a subset of y “.

The $x, y, z, \dots$ symbols are used as variables so their value can be a subset of a preliminary fixed set U .

With the help of the *Subset* language, compound terms cannot be created.

Examples of first-order languages

The Ar language

The *Ar* language is used to describe statements about nonnegative integers.

The $x, y, z, \dots$ symbols are variables (atomic terms) and arbitrary nonnegative integers can be assigned to them.

The constant symbol 0 (representing the integer zero) is also an atomic term, while compound terms can be constructed using s , $+$ and $\cdot$ where

sx can be read as “*the successor of x* ”,

$(x+y)$ can be read as “*the sum of x and y* ”,

$(x \cdot y)$ can be read as “*the multiplication of x and y* ”.

The atomic formulae of the *Ar* languages are in the form of $(t=z)$, meaning that the values of the terms t and z are equal.